REAL ESTATE ANALYTICS | MACHINE LEARNING | DATA ANALYTICS & VISUALIZATION | EXPLAINABLE AI

Explainable Real Estate Analytics: A Comparative Analysis of Machine Learning Frameworks for House Price Prediction

PROJECT REVIEW – 1

BATCH – 2

CH. Shalini — 2420030284

M. J. U. Harika — 2420030453

K. Aneesha — 2420030589

Faculty Guide: Dr. K Swapnika

Department: Computer Science and Engineering

University: KLH University

Academic Year: 2026 – 2027

FOUNDATION

Refined Title & Project Domain

REFINED TITLE

Explainable Real Estate Analytics: A Comparative Analysis of Machine Learning Frameworks for House Price Prediction

Why the title was refined

The title is refined from the initial comparative-analysis topic based on recent research in machine learning-based property valuation. The proposed work retains comparative evaluation and extends it by integrating feature engineering, ensemble learning and Explainable Artificial Intelligence (XAI).

PROJECT DOMAIN

Data Analytics & Visualization

Department: Computer Science and Engineering

University: KLH University

Faculty Guide: Dr. K Swapnika

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Problem Statement

House prices are influenced by multiple factors, making accurate prediction a complex task. This project presents a comparative machine learning framework using Decision Tree, Random Forest, XGBoost, and CatBoost, with comprehensive EDA, preprocessing, and feature engineering. Model performance is evaluated using MAE, MSE, RMSE, R^2, and MAPE, while SHAP and LIME provide interpretable insights into the key factors influencing house price predictions.

PROJECT AIM

To develop an explainable ensemble learning framework that improves house price prediction while identifying the major factors influencing the predicted price.

KEY INFLUENCING FACTORS

- **Location**
- **Property Area**
- **Number of Rooms**
- **Property Age**
- **Quality**
- **Amenities**

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Objectives

1 Data Preparation

To preprocess and prepare house-price data using missing-value treatment, categorical encoding, outlier handling and feature engineering techniques.

2 Model Implementation & Comparison

To implement and compare Linear Regression, Random Forest, XGBoost and CatBoost using K-Fold Cross-Validation and evaluation metrics such as MAE, RMSE and R^2.

3 Ensemble Development

To develop a Stacking Regressor by combining the best-performing machine learning models to improve house-price prediction performance.

4 Explainability

To apply SHAP-based Explainable Artificial Intelligence to identify and interpret the most influential features contributing to house-price predictions.

LITERATURE SURVEY

The following recent journal papers were reviewed to understand current machine learning approaches, ensemble methods, explainability techniques and research limitations in house-price prediction and property valuation.

Year	Author(s)	Paper	Method / Finding	Gap / Relevance
2026	Mammadzade, M.	Machine Learning Algorithms for House Price Prediction: A Comparative Study	Compares Linear Regression, Ridge Regression, Decision Tree and Random Forest for house-price prediction using the Ames Housing dataset.	Provides useful model comparison; scope remains for an integrated ensemble and XAI framework.
2025	Cao, Y., Xu, Y., Ji, Y., & Wang, H.	Automated Machine Learning for Residential Property Valuation	Proposes an AutoML framework for residential property valuation with automated model selection and optimization.	Strong automation, but relatively complex and not primarily focused on transparent prediction.
2025	Deng, L., & Zhang, X.	Boosting the Accuracy of Property Valuation with Ensemble Learning and Explainable Artificial Intelligence: The Case of Hong Kong	Uses ensemble learning for property valuation and XAI to identify important price determinants.	Highly relevant, but based on Hong Kong property data and a domain-specific valuation setting.
2025	Maselli, G., & Nesticò, A.	Machine Learning Algorithms and Explainable Artificial Intelligence for Property Valuation	Studies machine learning models for property valuation and applies SHAP and feature-importance techniques.	Shows the value of XAI; scope remains for combining comparative evaluation and stacking in one framework.

2025	Viktoratos, I., & Tsadiras, A.	Advancing Real-Estate Forecasting: A Novel Approach Using Kolmogorov–Arnold Networks	Investigates Kolmogorov–Arnold Networks for real-estate forecasting and compares them with other approaches.	Advanced model is more complex and does not primarily address interpretable ensemble prediction.
2025	Adonia, N., Mwangi, W. R., & Michael, K. W.	Optimizing Multivariate Adaptive Regression Splines with Coordinate Descent to Accurately Select the Best Model for House Price Prediction	Uses optimized Multivariate Adaptive Regression Splines with coordinate descent for house-price prediction and model selection.	Focuses on one optimized regression approach rather than complementary ensemble models with XAI.
2025	Karanikolas, N., Kyriakidou, E., & Athanasouli, E.	Artificial Intelligence and Real Estate Valuation: The Design and Implementation of a Multimodal Model	Develops a multimodal AI approach for real-estate valuation using different property information sources.	Multimodal modelling can require additional data and computational resources.
2025	Wang, X., & Kee, T.	Integrating Shapley Value and Least Core Attribution for Robust Explainable AI in Rent Prediction	Investigates Shapley-based attribution methods for explaining property rent predictions.	Strong focus on explainability, but the task is rent prediction rather than house-sale-price ensemble modelling.
2024	Jafary, P., Shojaei, D., Rajabifard, A., & Ngo, T.	Automating Property Valuation at the Macro Scale of Suburban Level: A Multi-Step Method Based on Spatial Imputation Techniques, Machine Learning and Deep Learning	Combines spatial imputation, machine learning and deep learning for macro-scale property valuation.	Powerful but spatially and computationally complex for a compact structured-data project.
2026	Kmen, C., Navratil, G., Kattenbeck, M., & Giannopoulos, G.	Evaluating Human–Machine Collaboration Through a Comparative Analysis of Experts, Machine Learning, and Hybrid Approaches in Real Estate Valuation	Compares machine learning, expert valuation and hybrid human–machine approaches using real-estate transaction data.	Focuses on human–machine collaboration rather than a standalone explainable ensemble framework.

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Research Gaps Identified

1. Many studies compare individual machine learning models, but there is scope for integrating complementary high-performing models into a single ensemble framework.
2. Although recent work combines ensemble learning and Explainable AI, several approaches are region-specific, spatially intensive or designed for specialized valuation environments.
3. Advanced AutoML, multimodal and deep-learning approaches can improve prediction but increase implementation and computational complexity.
4. Prediction accuracy is often emphasized more than the interpretability of individual predictions.
5. A practical framework combining feature engineering, model comparison, stacking and SHAP-based explanation remains a suitable research direction for structured house-price data.

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Novelty / Innovation

The novelty is not the individual use of ensemble learning or SHAP, since these techniques already exist in recent literature. The proposed innovation is their integration into a single practical house-price prediction pipeline.

1. Systematic feature engineering and preprocessing before model comparison.

2. Comparative evaluation of multiple regression models followed by selection of complementary high-performing models.

3. Use of a Stacking Regressor to combine selected models instead of relying only on one individual predictor.

4. Application of SHAP to the final model to provide understandable feature-level explanations for house-price predictions.

Feasibility

✓ Technical Feasibility:

- Implemented using Python and standard machine learning tools.
- Uses open-source libraries such as Scikit-learn, XGBoost, CatBoost, SHAP, and LIME.
- Public benchmark housing datasets are available for experimentation.
- Can be developed and executed on a standard computer.

✓ Economic Feasibility:

- No specialized paid software is required.
- Uses freely available datasets, tools, and open-source libraries.
- Requires minimal development and implementation costs.

✓ Operational Feasibility:

- Suitable for an academic machine learning and data analytics project.
- The workflow is structured, simple, and manageable.
- Can be completed within the planned project duration.
- Requires only standard computing resources.

METHODOLOGY

Proposed Workflow — Data Flow Pipeline

Data Collection → Preprocessing (Missing/Outlier/Encoding) → Feature Engineering → Train-Test Split (K-Fold) → Model Training (LR, RF, XGB, CB) → Stacking Regressor → SHAP Explainability → Comparative Evaluation

Preprocessed data flows through comparative model training, followed by ensemble stacking of the strongest performers, and finally SHAP-based explanation of the price drivers behind every prediction.

EXECUTION

Project Plan & Timeline

Phase 1	Foundational groundwork and literature review (planned activity to be detailed as the project progresses).
Phase 2	Feature engineering and exploratory data analysis; prepare relevant features for model development.
Phase 3	Comparative implementation of Linear Regression, Random Forest, XGBoost and CatBoost (planned activity to be detailed as the project progresses).
Phase 4	Develop and evaluate the Stacking Regressor; compare ensemble performance with individual models; perform SHAP-based explainability and feature analysis.
Phase 5	Testing and final evaluation; analyze results; prepare documentation, report and presentation.

EXECUTION

Team Details

1

CH. Shalini

Roll No. 2420030284

Literature Survey, Data Preprocessing & Feature Engineering

2

M. J. U. Harika

Roll No. 2420030453

ML Model Development, Model Comparison & Ensemble Learning

3

K. Aneesha

Roll No. 2420030589

SHAP Analysis, Result Analysis, Documentation & Presentation

Thank You

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